SANKUR KUNDU

ML Systems Engineer & Independent Researcher | LLM Training, Inference & GPU Kernels

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SUMMARY

ML systems engineer and independent researcher (B.Tech CSE, 2026): sole author of **four manuscripts**, merged fixes in **Meta ExecuTorch** and **Hugging Face OpenEnv**, PRs in review at **Google JAX**, **NVIDIA**, **Apple MLX**, **vLLM**, **PyTorch** and research labs at **MIT** and **Oxford, UCL & Imperial**. I build ML systems from scratch against frontier baselines: a tokenizer beating **GPT-4o's and Gemini's** in 23 of 24 Indian languages, RL post-training (GSM8K **48.1% → 52.5%**), kernels at **cuBLAS parity**, a **GPU quantum decoder** **bit-identical to IBM's**, and a **collision-risk engine** matching **NASA** on **53/53** cases (now under review by NASA CARA).

RESEARCH AND RELEASES

- Sole author: **"Removing the Token Tax from an Existing Model"**, **"The Sampler Was Not the Policy"** and **"Reproducing NASA CARA's Monte Carlo Probability of Collision"** (arXiv, forthcoming); "Scam Defence Stops at English" (in preparation).

RECOGNITION AND OPEN SOURCE

- **NASA CARA**: my reproduction of its Monte Carlo results (**53/53** test cases) is under review by its analysts.
- **Entrepreneur First** (UK): scouted from my public work alone by its Talent Investor, citing my **cuBLAS-parity** compiler.
- **Meta ExecuTorch #23677** (**merged**), **#23912** (in review): fixed out-of-bounds reads in on-device LLM kernels.
- **Hugging Face OpenEnv #1441**, **#1442** (**both merged**): fixed hook order and async scoring in RL reward rubrics; **9** new tests.
- In review: **Google JAX**, **NVIDIA**, **Apple MLX**, **vLLM**, **PyTorch**, **MIT Han Lab**, **Oxford PyZX**, **BOLD Lab** (Oxford, UCL, Imperial).

MACHINE LEARNING PROJECTS

ratchet — Python, PyTorch and C++/CUDA: reinforcement-learning post-training (GRPO) from scratch · study page

- Built GRPO post-training end to end: rollouts with per-token log-probabilities on relay (through a C API), a PyTorch trainer data-parallel on tandem, in-place GPU weight sync, partial rollouts and one-step-ahead training.
- On 2× T4, 100 steps lifted Qwen2.5-0.5B on GSM8K from **48.1% to 52.5%** (all 1,319 test problems); weight sync is **369×** faster than a reload, and overlapping generation with training makes steps **1.9×** faster.
- Found that the engine's fp16 weights rounded away most updates, so it sampled a policy the trainer never had (gap up to 5.2 nats); mixed-precision rounding cut it to **0.04**, and sync rounding outweighs fp16 arithmetic **9–22×**.

lipi — Python and PyTorch: cutting the token tax on Indian languages · study page

- Measured the token tax of 24 Indian languages on 15 tokenizers (Odia: **4.99×** English on GPT-4o); built **lipi-Indic** (**on Hugging Face**): fewer tokens than **GPT-4o's and Gemini's** in **23 of 24** languages (**1.30×** vs 1.92×, 1.58×).
- Extended Qwen2.5-0.5B's vocabulary and continued pre-training on 2× T4 (DDP, ZeRO-1): Odia takes **8.6× fewer** tokens, and in equal GPU time the model predicts held-out Odia **20% better**, with Odia→English chrF++ 16.0 → 24.8.

babelguard — Python: an open early-warning system for scams that cross languages (in progress) · study page

- Measured the protection gap: an English keyword filter catches **86%** of English scams but **10–36%** of Korean, Japanese, Urdu, Arabic, Thai and Hindi ones.
- Built linking by meaning that recognises a translated scam **85%** of the time (text matching: **1.5%**) and a crash-safe streaming engine: **4,057 messages/s** on 4 cores, **0.21 KB** per message at 1M, nothing lost in **50** kills.

Additional projects

- **relay**: fault-tolerant disaggregated LLM serving (C++/CUDA/Swift): prefill **8.7×** faster on custom kernels; **384/384** requests identical through 60 worker kills.
- **crucible**: sandboxes for agent RL forked in **4.5 ms**; GRPO graded in them lifted MBPP **34.9% → 38.6%** (3 seeds, p = 0.013).
- **tessel**: GEMM at **cuBLAS parity**, LLM engine **1.05×** vLLM; **kiln**: beats **torch.compile, ONNX Runtime, XLA** on CPU.
- **switchyard**: MoE serving, **2.2×** Hugging Face, dispatch **1.25×** faster than NCCL; **tandem**: training **bit-identical** to PyTorch.

EXPERIENCE

Gigzen — Co-founder & Founding Engineer **Apr 2026 – Present**

Early-stage startup (company name approved by India's MCA; incorporation in progress) · Sole engineer across every product

- Sole engineer on **Waggle**, three Android apps on Supabase PostgreSQL: **53 tables, 90 RLS policies, 154 SQL functions**.
- Built **Populace**, an open-source testing tool on **npm** (**1,600+ downloads**) whose first run found **5 bugs** manual QA missed.

SKILLS

ML systems: GRPO/PPO, DDP, ZeRO-1/2/3, FSDP, 1F1B, MoE expert parallelism, disaggregated serving, paged KV, BPE tokenizers; PyTorch, Hugging Face

GPU & languages: CUDA (tensor cores, streams, peer-to-peer), Triton-style tile languages, Pallas (TPU), Metal, ONNX; Python, C++, Rust, Swift, SQL

EDUCATION

B.Tech, Computer Science and Engineering (AI & ML) — IEM Kolkata **Sep 2022 – Apr 2026**